europass

Rafli Rizaldi

Home address: (Indonesia)

Skills

Programming and data analysis:

- Python (computer programming)
- SQL
- ADQL
- Git / GitHub
- Jupyter
- LaTeX

Astronomical data and tools:

- Astrophysics software: SAOImage, DS9, IRAF, TOPCAT, SM
- Gaia Archive
- ASteCA
- Parsec isochrones
- galpy
- lenstronomy
- SDSS CasJobs

Research methods:

- statistics
- Bayesian inference
- machine learning
- Scientific writing and presentation

Language Skills

- **Indonesian**: Mother tongue
- **English**
  - Listening: C1
  - Reading: C2
  - Spoken production: B2
  - Spoken interaction: B2
  - Writing: B2

Education & Training

Master of Science in Astronomy
01/09/2025 - Current

Institut Teknologi Bandung (ITB) | Bandung, Indonesia

Thesis: Kinematic Signatures of Rotation and Expansion in Open Cluster Tidal Tails

Supervisors: Dr. rer. nat. M. Ikbal Arifyanto and Dr. M. Irfan Hakim

GPA: 3.93/4.00

Main courses: Stellar Physics, Theory and Observation of Stars, Physics of Galaxies, Topics in Computational Astrophysics, Capita Selecta in Galaxy and Cosmology, Astronomy Laboratory, Research Methodology

Level in EQF: 7

International School for Young Astronomers (ISYA)
28/09/2026 - 16/10/2026

International Astronomical Union (IAU) – Office for Young Astronomers, and ICISE | Quy Nhon, Vietnam

Three-week intensive school on theoretical and observational astrophysics, with lectures on stellar evolution, galaxies, cosmology, the interstellar medium, exoplanets, radio and optical astronomy, virtual observatory, and machine learning; hands-on data-analysis tutorials and a research project using observational and simulation data; career development workshop.

Bachelor of Science in Astronomy
16/08/2021 - 29/07/2025

Institut Teknologi Bandung (ITB) | Bandung, Indonesia

Thesis: Determination of Membership and Physical Parameters of Open Clusters of the Solar Neighborhood Using kNN and GMM

Supervisor: Dr. M. Irfan Hakim

GPA: 3.80/4.00, Magna Cum Laude

Main courses: Mechanics, Electromagnetic Fields, Mathematical Methods in Astronomy I & II, Statistics and Data Mining, Computational Astronomy, Astrophysics I & II, Stellar Physics, Stellar System, Processes in Astrophysics I & II, Astrophysical Hydrodynamics, Physics of Galaxy, Observational Astrophysics

Level in EQF: 6

Projects

Rotation and Expansion in Open Cluster Tidal Tails (MSc thesis)
01/09/2025 - Current

Supervisors: Dr. rer. nat. M. Ikbal Arifyanto, Dr. M. Irfan Hakim. Measured rotation and expansion/contraction in the cores and tidal tails of 19 open clusters using Gaia DR3. Presented at MODEST-26; paper in preparation.

INAGLENS – Strong Gravitational Lens Modelling
18/04/2024 - Current

Supervisors: Dr. Anton T. Jaelani, Dr. Anowar J. Shajib, Dr. Kenneth Wong. Modelling strong gravitational lenses in Subaru HSC imaging with lenstronomy.

Dust Attenuation and Star-Formation Timescales in Nearby Galaxies
23/07/2025 - 14/10/2025

Supervisor: Dr. Dian P. Triani. Testing whether the high-redshift AV–sSFR relation holds for local SDSS galaxies. Presented as a first-author poster at the 15th SEAAN Meeting.

Parallax-Based GMM for Open Cluster Membership (BSc thesis)
01/07/2024 - 29/07/2025

Supervisor: Dr. M. Irfan Hakim. Developed a parallax-based GMM extension of the kNN+GMM membership method for poorly studied open clusters. Published in J. Phys.: Conf. Ser. 3234, 012015 (2026).

Publications

Kinematic signatures of rotation and expansion in open cluster tidal tails

R. Rizaldi, M. I. Arifyanto, M. I. Hakim

First-author paper in preparation; results presented as an oral talk at MODEST-26 (2026).

Rizaldi, R., Arifyanto, M. I., Hakim, M. I. (in prep.). Kinematic signatures of rotation and expansion in open cluster tidal tails.

A parallax distribution based Gaussian Mixture Model for membership determination in open clusters of solar neighborhood

R. Rizaldi, M. I. Hakim, B. Randolph (2026) | 3234, 1, 012015 | Journal of Physics: Conference Series | IOP Publishing

Rizaldi, R., Hakim, M. I., Randolph, B. (2026). A parallax distribution based Gaussian Mixture Model for membership determination in open clusters of solar neighborhood. Journal of Physics: Conference Series, 3234, 012015.

https://doi.org/10.1088/1742-6596/3234/1/012015

Conferences & Seminars

International Teeminar (Stellar Dynamics Group) 15/12/2026 - 15/12/2026

Online

Invited online teeminar talk (scheduled): "Rotation, Expansion, and Contraction in the Tidal Tails of Open Clusters". Hosted by Prof. Dr. Rainer Spurzem.

https://astro-silkroad.eu/seminars/teeminar-heidelberg

MODEST-26 International Conference 16/08/2026 - 22/08/2026

Quy Nhon, Gia Lai, Vietnam

Oral talk: "Kinematic signatures of rotation and expansion in open cluster tidal tails". International conference on observing and modelling dense star clusters.

https://www.icisequynhon.com/conferences/2026/modest2026/program.html

15th Southeast Asia Astronomy Network (SEAAN) Meeting 13/10/2025 - 15/10/2025

Bandung, Indonesia

Oral presentation (first author): "A parallax distribution based Gaussian Mixture Model for membership determination in open clusters of solar neighborhood"; and poster (first author): "The AV–1/sSFR relation in nearby galaxies".

Honours and Awards

Ganesha Talent Assistantship 01/09/2025

Institut Teknologi Bandung (ITB)

Graduate scholarship providing full tuition support in exchange for part-time academic and research assistance.

Finalist, Pre-9th Mission Idea Contest: to the Moon (PreMIC9) 27/11/2024

University Space Engineering Consortium (UNISEC-Global)

Led a nine-member ITB team proposing "Investigating the Effects of Cosmic Radiation on Human Cells in Lunar Orbit", a lunar-orbit CubeSat mission. Selected as one of the finalists and presented the concept at the final presentation in Stellenbosch, South Africa.

https://www.spacemic.net/index9pre.html

POSCO Asia Fellowship for Asian Universities 31/10/2024

POSCO TJ Park Foundation

Scholarship for outstanding undergraduate students at Asian universities.

Dean's List 01/08/2024

Faculty of Mathematics and Natural Sciences, ITB

Awarded for a perfect 4.00 GPA

Work experience

Teaching Assistant – Topics in Computational Astrophysics (MSc course) 01/09/2025 - Current

Institut Teknologi Bandung | Bandung, Indonesia

- Led tutorials on Bayesian distances from parallaxes, Monte Carlo simulations, the IMF, synthetic clusters, and clustering methods for open clusters.
- Supported practical exercises.

Technical Team Member 01/02/2025 - 21/09/2026

Astronomy Olympiad Coaching Team (TPOA), Institut Teknologi Bandung | Bandung, Indonesia

- Trained the Indonesian national team for the International Olympiad on Astronomy and Astrophysics (IOAA), focusing on the observational round.
- Supported national training camps for the Indonesian astronomy olympiad team.
- Handled instrumentation and observation components of the National Science Olympiad (OSN) in astronomy, including jury support.

Teaching Assistant – Computational Astronomy 01/09/2025 - 31/01/2026

Institut Teknologi Bandung | Bandung, Indonesia

- Led tutorials on error analysis, root finding, linear systems, interpolation, regression, numerical integration, and ODEs.
- Supervised computational lab sessions.

Teaching Assistant – Astrophysics I & II 01/01/2024 - 31/12/2025

Institut Teknologi Bandung | Bandung, Indonesia

- Covered stellar photometry and spectroscopy, stellar structure and evolution, radiation laws, binary stars, and introductory cosmology.
- Facilitated discussions and problem sets.

Teaching Assistant – Positional Astronomy 01/08/2023 - 31/12/2023

Institut Teknologi Bandung | Bandung, Indonesia

- Led tutorials on celestial coordinate and time systems, precession, nutation, and apparent motions of the Sun and Moon.

Lab Assistant – Elementary Physics Laboratory 01/01/2023 - 31/05/2023

Institut Teknologi Bandung | Bandung, Indonesia

- Supervised laboratory sessions on laser optics, interferometry, and basic experimental physics.